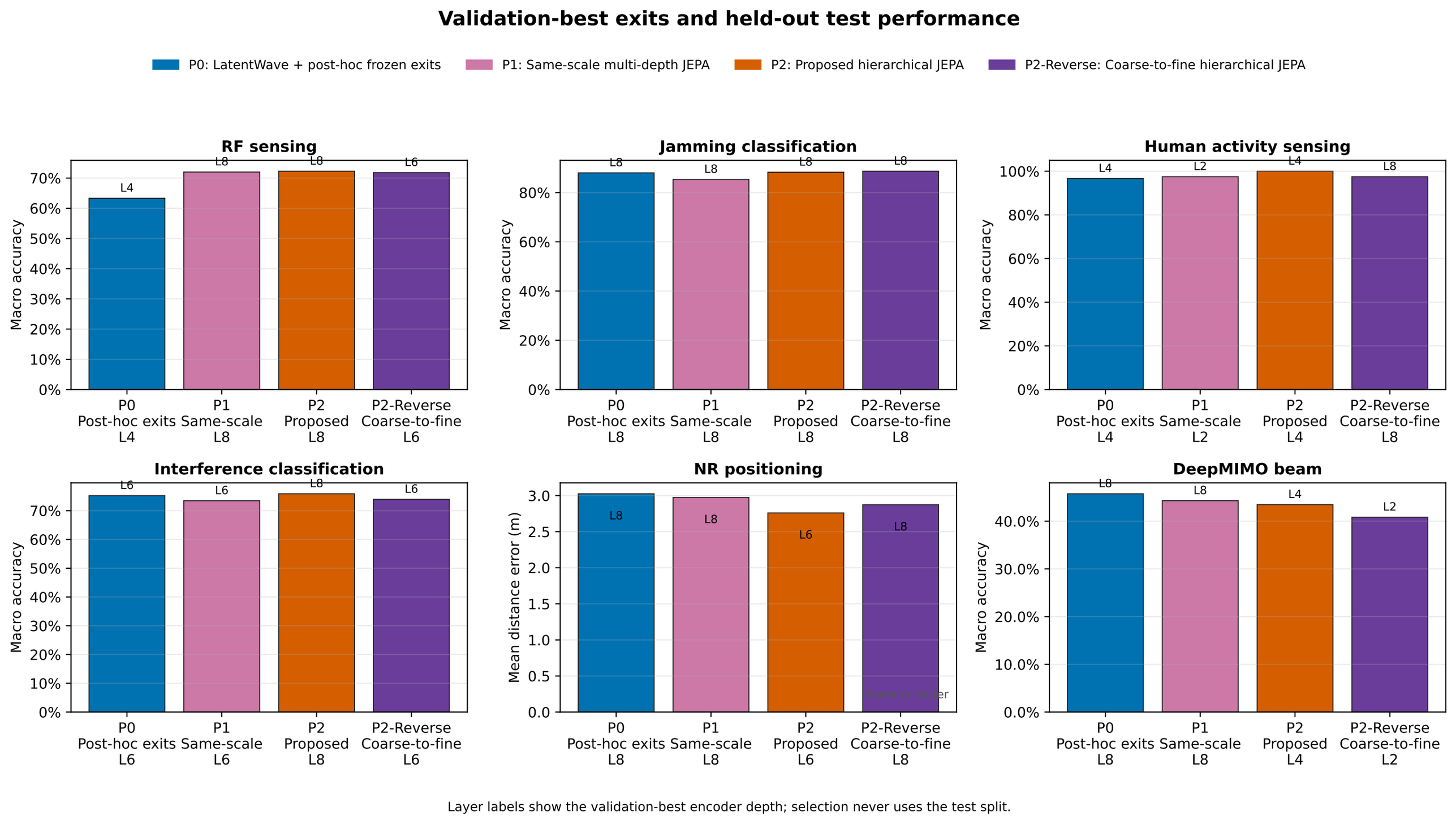
Validation-best exits and held-out test performance
P0: LatentWave + post-hoc frozen exits
P1: Same-scale multi-depth JEPA
P2: Proposed hierarchical JEPA
P2-Reverse: Coarse-to-fine hierarchical JEPA
RF sensing
Macro accuracy
L4
L8
L8
L6
P0 Post-hoc exits L4
P1 Same-scale L8
P2 Proposed L8
P2-Reverse Coarse-to-fine L6
Jamming classification
Macro accuracy
L8
L8
L8
L8
P0 Post-hoc exits L8
P1 Same-scale L8
P2 Proposed L8
P2-Reverse Coarse-to-fine L8
Human activity sensing
Macro accuracy
L4
L2
L4
L8
P0 Post-hoc exits L4
P1 Same-scale L2
P2 Proposed L4
P2-Reverse Coarse-to-fine L8
Interference classification
Macro accuracy
L6
L6
L8
L6
P0 Post-hoc exits L6
P1 Same-scale L6
P2 Proposed L8
P2-Reverse Coarse-to-fine L6
NR positioning
Mean distance error (m)
lower is better
L8
L8
L6
L8
P0 Post-hoc exits L8
P1 Same-scale L8
P2 Proposed L6
P2-Reverse Coarse-to-fine L8
DeepMIMO beam
Macro accuracy
L8
L8
L4
L2
P0 Post-hoc exits L8
P1 Same-scale L8
P2 Proposed L4
P2-Reverse Coarse-to-fine L2
Layer labels show the validation-best encoder depth; selection never uses the test split.